

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	Hard

Question

The density of a certain solid substance is **813** kilograms per cubic meter. A sample of this substance is in the shape of a cube, where each edge has a length of **0.60** meters. To the nearest whole number, what is the mass, in kilograms, of this sample?

Answer

- A. **176**
- B. **488**
- C. **1,355**
- D. **3,764**

Correct Answer: A

Rationale

Choice A is correct. It's given that the sample is in the shape of a cube with edge lengths of **0.60** meters. Therefore, the volume of the sample is **0.60³**, or **0.216**, cubic meters. It's also given that the sample has a density of **813** kilograms per **1** cubic meter. Therefore, the mass of this sample is **(0.216 cubic meters) ($\frac{813 \text{ kilograms}}{1 \text{ cubic meter}}$)**, or **175.608** kilograms. Rounding this mass to the nearest whole number gives **176** kilograms. Therefore, to the nearest whole number, the mass, in kilograms, of this sample is **176**.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.